

THE UNIVERSITY OF HONG KONG
DEPARTMENT OF MATHEMATICS

ARIN7013: NUMERICAL METHODS IN ARTIFICIAL INTELLIGENCE

May 15, 2023

6:30p.m. – 9:00p.m.

Only approved calculators as announced by the Examinations Secretary can be used in this examination. It is candidates' responsibility to ensure that their calculator operate satisfactorily, and candidates must record the name and type of the calculator used on the front page of the examination script.

Answer ALL 4 questions

Read each question carefully. You may not be credited any score if you put down your conclusion ONLY.

Instructions

- Please include the calculations in your solution neatly. This requirement is to ensure that you are not using any mathematics software (which is forbidden).
- You should stop working at 9:00pm.
- Your cheat sheet should be handwritten, double sided A4 paper, otherwise, your cheat sheet will be taken away.

Note: You should always give precise and adequate explanations to support your conclusions. Clarity of presentation of your argument counts. So **think carefully before you write.**

1. (20 points) Let $x = \{x_1, \dots, x_n\}$ be a subset of an input data $\mathcal{X}$. Consider a positive definite kernel $k : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$, which induces a feature map $\psi : \mathcal{X} \rightarrow (\mathcal{H}, \|\cdot\|_{\mathcal{H}})$. Here, $(\mathcal{H}, \|\cdot\|_{\mathcal{H}})$ denotes the associated Reproducing Kernel Hilbert Space (RKHS).

- (a) (5 points) Deduce an expression, which allows to calculate the average distance to the center of mass of the image of set x in the feature space:

$$\frac{1}{n} \sum_{i=1}^n \|\psi(x_i) - \psi_S(x)\|_{\mathcal{H}},$$

where the center of mass is defined as

$$\psi_S(x) = \frac{1}{n} \sum_{i=1}^n \psi(x_i).$$

- (b) (5 points) Is

$$k(x, y) = e^{-\frac{\|x-y\|_{\ell^2}^2}{2}} \text{ for } x, y \in \mathbb{R}^2$$

a positive definite kernel? If so, what is the corresponding RKHS?

- (c) (10 points) Use the expression in (a) to calculate the average distance to the center of mass of the following point set in $\mathbb{R}^2$,

$$x = \{(0, 1), (-1, 3), (2, 4), (3, -1), (-1, -2)\},$$

in the RKHS induced by the following kernels,

- i. $k(x, y) = \langle x, y \rangle_{\ell^2}$
- ii. $k(x, y) = \langle x, y \rangle_{\ell^2}^2$
- iii. $k(x, y) = (\langle x, y \rangle_{\ell^2} + 1)^5$

2. (25 points) Power iteration is designed to compute the dominant eigenvalue of a matrix $A \in \mathbb{R}^{m \times m}$ and its associated eigenvector. Let $A \in \mathbb{R}^{m \times m}$ be diagonalizable with eigenvalues

$$|\lambda_1| > |\lambda_2| \geq |\lambda_3| \geq \cdots \geq |\lambda_m|$$

and normalized eigenvectors $\{u_1, \dots, u_m\}$. The procedure goes as the following: pick an arbitrary initial vector $x_0 \in \mathbb{C}^m$, generate sequences

$$\begin{aligned} z_k &= Ax_k, \\ x_{k+1} &= \frac{z_k}{\|z_k\|_2}, \\ r_{k+1} &= x_{k+1}^T Ax_{k+1}. \end{aligned}$$

- (a) (5 points) Suppose the initial vector x_0 has an expansion of the form $x_0 = a_1 u_1 + \cdots + a_m u_m$ with $a_1 \neq 0$. Use this expansion of x_0 to write $A^k x_0$ into the following form and find the exact expression for ε_k

$$A^k x_0 =: a_1 \lambda_1^k (u_1 + \varepsilon_k) \text{ for } k \geq 1.$$

- (b) (5 points) Prove that if the initial vector x_0 has an expansion of the form $x_0 = a_1 u_1 + \cdots + a_m u_m$ with $a_1 \neq 0$, then as $k \rightarrow \infty$, x_{k+1} aligns along with direction of u_1 , and

$$r_k \rightarrow \lambda_1 \text{ as } k \rightarrow \infty$$

You may use the result of (a).

- (c) (5 points) Let $x_0 = a_1 u_1 + \cdots + a_m u_m$ be any vector with $a_1 \neq 0$. Show that there is a constant $c > 0$, such that

$$\|y_k - u_1\|_2 \leq c \left| \frac{\lambda_2}{\lambda_1} \right|^k \quad \text{for } y_k = \frac{x_k \|A^k x_0\|_2}{a_1 \lambda_1^k}$$

This implies that if $|\lambda_2|$ is close to $|\lambda_1|$, then the convergence of sequence $\{y_k\}_k$ to eigenvector u_1 is slow.

- (d) (5 points) The inverse iteration algorithm is mainly based on the following result: if λ is an eigenvalue of A and if A is invertible, then $\frac{1}{\lambda}$ is an eigenvalue of A^{-1} . Please prove it.

- (e) (5 points) The Rayleigh quotient iteration algorithm combines the idea of Rayleigh quotient and inverse iteration. Please formulate the Rayleigh quotient iteration algorithm. What's its convergence rate (state without proof)?

3. (30 points) Given three data points in $\mathbb{R}^3$: $x_1 = [1, 2, 3]^\top$, $x_2 = [3, 4, 5]^\top$, $x_3 = [2, 3, 1]^\top$.

- (a)
 - i. (2 points) Find the mean m of these three points and the corresponding x_1, x_2, x_3 centered with respect to m .
 - ii. (2 points) Find the covariance matrix of x_1, x_2, x_3 .
 - iii. (5 points) What's the eigenvalues and eigenvectors of this covariance matrix?
 - iv. (5 points) What's the first principal component of these three data points? The first principal component carries how many percentage of the information (correct your answer into 1 decimal digit)?
 - v. (5 points) What's the second principal component of these three data points? How much information can be contained by using the first two principal components (correct your answer into 1 decimal digit)?

(b) Let $k(x, y) = \langle x, y \rangle_{\ell^2}^2$ for $x, y \in \mathbb{R}^3$. Let $\mathcal{H}$ be the corresponding RKHS for the kernel $k(\cdot, \cdot)$, and let $\Psi : \mathbb{R}^3 \rightarrow \mathcal{H}$ be the kernel map. We plan to apply kernel PCA to find the important modes from the same data points, i.e., we want to apply PCA to the following set of functions in $\mathcal{H}$,

$$\{\Psi(x_1), \Psi(x_2), \Psi(x_3)\}.$$

- i. (3 points) Calculate the gram matrix.
- ii. (5 points) Solve the eigenvalue problem of this gram matrix.
- iii. (3 points) Let $x = [1, 1, 1]^\top$. What's the projection of x onto the first principal component of this kernel PCA?

4. (25 points)

(a) (15 points) Let

$$\frac{dx}{dt} = 2xt, \quad x(0) = 3.$$

- i. (5 points) What is the exact solution $x(t)$?
- ii. (5 points) Use Forward Euler scheme with mesh size $h = 0.25$ to calculate the numerical solution $x(ih)$ for $i = 1, 2, 3$.
- iii. (5 points) Use Backward Euler scheme with mesh size $h = 0.25$ to calculate the numerical solution $x(ih)$ for $i = 1, 2, 3$.

(b) (10 points) Neural ODE can be applied to build the following model,

$$\frac{dx}{dt} = f(x(t), t, \theta), \quad x(0) \text{ is given .}$$

Here, the source term $f(x(t), t, \theta)$ is modeled by a neural network with θ being the associated parameters to be trained by the training data.

- i. (3 points) Where can we get the training data?
- ii. (7 points) Let $L(\cdot, \cdot)$ be the loss function. Can you use the adjoint method to calculate $\frac{\partial L}{\partial \theta}$?

***** END OF PAPER *****